Food Nutrition and diet Recommendation System

SQL DATABASE PROJECT

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🥗 Objective of the Project

The main objective of the Food Nutrition Database Project is to design a structured SQL database that stores and manages users' dietary and nutritional information efficiently. It aims to:

1. Track users' personal health details such as age, weight, height, and activity level.

2. Store detailed food data including calories, protein, carbohydrates, and fat content.
3. Link users to specific diet plans and recommended meals.
4. Generate meaningful insights using SQL queries such as JOIN, subqueries, and aggregate functions.
5. Help users and administrators make better diet-related decisions through organized data analysis.

Table Structures

Example of table structures used in the project:

- **Users Table Columns**: user_id, name, age, gender, height_cm, weight_kg, activity_level, email.
- **Food Items Table Columns**: food_id, food_name, category, calories, protein_g, carbs_g, fat_g, user_id.
- **Diet Plan Table Columns**: plan_id, plan_name, goal, duration_weeks.
- **Diet Recommendations Table Columns**: rec_id, user_id, food_id, plan_id, meal_time, portion_size.

SQL Queries

1.WAQTD all food items in the "Proteins" category

```
select *
from food_items
where category='Protein';
```

food_id	food_name	category	calories	protein_g	carbs_g	fat_g	user_id
3	Boiled Egg	Protein	78	6.30	0.60	5.30	3
5	Grilled Chicken Breast (100g)	Protein	165	31.00	0.00	3.60	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2.Count how many foods belong to each category

```
select category,count(*)  as total_items
from  food_items
group by category;
```

category	total_items
Fruit	2
Protein	2
Grain	2
Vegetable	1
Nuts	1
Dairy	2

3.Shows top 3 high-calorie foods

```
select food_name,calories
from food_items
order by calories desclimit 3;
```

food_name	calories
Paneer (100g)	265
Brown Rice (1 cup)	216
Grilled Chicken Breast (100g)	165

4. Shows 5 youngest users

```
select name,age
 from users
order by age asc
limit 5;
```

name	age
Neha Desai	22
Isha Nair	24
Aarav Mehta	25
Rahul Verma	27
Rohan Patel	28

5.Shows all food items that belong to the 'Dairy' category'

```
select food_name,category
from food_items
where category='Dairy';
```

food_name	category
Milk (1 glass)	Dairy
Paneer (100g)	Dairy

6. all female users older than 25

```
select name,age,gender
from users
where gender='Female' and age>25;
```

name	age	gender
Priya Sharma	30	Female
Ananya Joshi	35	Female
Sneha Kapoor	29	Female

7.Find total calories in each food category

```
SELECT category, SUM(calories) AS
total_calories
FROM food_items
GROUP BY category
ORDER BY total_calories DESC;
```

category	total_calories
Dairy	415
Grain	366
Protein	243
Fruit	200
Nuts	70
Vegetable	55

8.Shows food categories having total calories greater than 200

```
SELECT category, SUM(calories) AS total_calories
FROM food_items
GROUP BY category
HAVING SUM(calories) > 200;
```

category	total_calories
Protein	243
Grain	366
Dairy	415

9.Find all users older than the average age of users

```
select name,age
from users
where age >(select avg(age)
 from users);
```

name	age
Priya Sharma	30
Ananya Joshi	35
Sneha Kapoor	29
Vikram Singh	33
Karan Bansal	31

10. Display all food items added by users with 'High' activity level

```
SELECT food_name
FROM food_items
WHERE user_id IN (SELECT user_id
FROM users
   WHERE activity_level = 'High');
```

food_name
Boiled Egg
Grilled Chicken Breast (100g)

11.List users who have been recommended at least one food from the 'Protein' category

```
SELECT DISTINCT name
FROM users
WHERE user_id IN (SELECT user_id
 FROM diet_recommendations
WHERE food_id IN (SELECT food_id
 FROM food_items
  WHERE category = 'Protein'));
```

name
Rohan Patel

12. Shows all foods recommended to users younger than 30

```
SELECT food_name
FROM food_items
WHERE food_id IN (SELECT food_id
FROM diet_recommendations
 WHERE user_id IN (SELECT user_id
FROM users
WHERE age < 30));
```

food_name
Apple
Boiled Egg
Brown Rice (1 cup)
Grilled Chicken Breast (100g)
Broccoli (1 cup)
Paneer (100g)

13. Find all users who are following the same diet plan as 'Aarav Mehta’

```
SELECT DISTINCT u.name, d.plan_name
FROM users u
JOIN diet_recommendations dr ON u.user_id = dr.user_id
JOIN diet_plan d ON dr.plan_id = d.plan_id
WHERE dr.plan_id IN ( SELECT plan_id
 FROM diet_recommendations
 WHERE user_id = ( SELECT user_id
FROM users WHERE name = 'Aarav Mehta'));
```

	name	plan_name
▶	Aarav Mehta	Weight Loss

14. Find diet plans that include the same foods as the "Balanced Diet" plan

```
SELECT DISTINCT p.plan_name
FROM diet_plan p
JOIN diet_recommendations d ON p.plan_id = d.plan_id
WHERE d.food_id IN ( SELECT food_id
 FROM diet_recommendations
 WHERE plan_id = (SELECT plan_id
FROM diet_plan
WHERE plan_name = 'Balanced Diet'));
```

	plan_name
▶	Balanced Diet

15. Shows users whose total recommendede food calories are above the average total calories

```
SELECT u.name, SUM(f.calories) AS total_calories
FROM users u
JOIN diet_recommendations d ON u.user_id = d.user_id
JOIN food_items f ON d.food_id = f.food_id
GROUP BY u.name
HAVING total_calories > (SELECT AVG(total_calories)
  FROM (SELECT SUM(f2.calories) AS total_calories
  FROM diet_recommendations d2
 JOIN food_items f2 ON d2.food_id = f2.food_id
 GROUP BY d2.user_id    ) AS avg_calories);
```

	name	total_calories
▶	Aarav Mehta	311
	Neha Desai	320

🍽️ Conclusion

- The Food Nutrition and Diet Recommendation System Database effectively organizes and manages nutritional information for users.
- It helps track individual diet plans, food intake, and nutritional values using SQL relationships and queries.
- By implementing features like JOINs, subqueries, and constraints, this project demonstrates how databases can be used to create a health-focused, data-driven system